TYPHOID FEVER

Custom Study Summary with reference to National Health Portal (NHP) and Government public-health guidance

Overview

Typhoid fever is a serious systemic bacterial infection caused by *Salmonella enterica* serovar Typhi, commonly known as *Salmonella Typhi*. It is mainly transmitted through ingestion of food or water contaminated with human faecal matter containing the bacteria. The disease is closely associated with inadequate sanitation, unsafe drinking water and poor food hygiene.

Key Facts

• Causative agent: *Salmonella Typhi*.

• Type of disease: Bacterial systemic infection, also known as enteric fever.

• Reservoir: Humans are the only natural reservoir.

• Main mode of spread: Contaminated food and drinking water.

• Incubation period: Symptoms commonly develop about 6–30 days after exposure.

• Untreated infection may lead to serious intestinal and systemic complications.

Transmission

The usual transmission cycle is:

Infected person/carrier → bacteria shed in faeces → contamination of food or water → ingestion by another person → Typhoid infection

• Infected individuals may contaminate food or water through inadequate hand hygiene.

• Chronic carriers can continue to shed bacteria even after recovery.

• Poor sanitation and unsafe water supplies increase the risk of community transmission.

Clinical Features and Symptoms

Typhoid often has an insidious onset and may initially resemble other febrile illnesses. Common symptoms include:

• Prolonged or sustained fever

• Headache

• Fatigue and weakness

• Loss of appetite

• Nausea

• Abdominal pain

• Constipation or diarrhoea

• Dry cough in some patients

• Rose-coloured spots or rash in some cases

Clinical Progression

The illness may progress gradually over several days. Persistent fever and systemic symptoms can be accompanied by gastrointestinal manifestations. Without appropriate treatment, prolonged infection may

increase the risk of complications.

Complications

Severe or untreated typhoid can cause serious complications, including:

• Intestinal bleeding

• Intestinal perforation and peritonitis

• Severe systemic toxicity

• Encephalopathy or altered mental status

• Myocarditis or other organ involvement

Diagnosis

Diagnosis should be based on clinical assessment together with laboratory confirmation. Diagnostic approaches may include:

• Blood culture, particularly during appropriate stages of illness

• Culture of other relevant clinical specimens where indicated

• Antigen or serological tests as supportive methods

• Antimicrobial susceptibility testing to guide antibiotic selection

Treatment and Management

Typhoid fever is treated with appropriate antibiotics prescribed by a qualified healthcare professional. Antibiotic resistance can affect treatment choices, making laboratory confirmation and local treatment guidance important.

• Seek medical assessment for prolonged fever or suspected typhoid.

• Complete the entire prescribed antibiotic course.

• Maintain adequate hydration and nutritional support as advised.

• Do not self-medicate with antibiotics.

• Follow-up may be required to identify persistent carriage or treatment failure.

Prevention and Control

Prevention primarily depends on safe water, sanitation, food hygiene and vaccination where recommended.

Safe food and water:

• Drink safe, treated or boiled water.

• Eat freshly cooked and hygienically prepared food.

• Wash fruits and vegetables with safe water.

• Avoid food and beverages from unhygienic sources.

Personal and community hygiene:

• Wash hands thoroughly with soap, especially after using the toilet and before handling food.

• Use sanitary toilets and ensure safe disposal of human waste.

• Promote hygiene among food handlers.

• Support improvements in sanitation and safe water supply.

Vaccination:

• Typhoid vaccines are available and may be recommended for people at risk according to public-health guidance.

• Vaccination complements, but does not replace, safe food, water and hygiene practices.

When to Seek Medical Attention

• Persistent or prolonged fever

• Severe abdominal pain

• Persistent vomiting or inability to maintain fluids

• Blood in stool or black stools

• Confusion, extreme weakness or altered consciousness

• Rapid worsening of symptoms

Typhoid: Quick Summary

Cause: *Salmonella Typhi*
Spread: Contaminated food and water via the faecal-oral route
Incubation: Commonly 6–30 days
Main symptoms: Prolonged fever, headache, weakness, abdominal symptoms and altered bowel habits
Diagnosis: Clinical assessment with laboratory testing, especially culture where appropriate
Treatment: Appropriate prescribed antibiotics and supportive care
Prevention: Safe water, sanitation, hand hygiene, food safety and vaccination
Complications: Intestinal bleeding/perforation and severe systemic illness in untreated cases

References: National Health Portal (NHP), Ministry of Health & Family Welfare, Government of India; Government of India public-health resources on enteric fever and standard treatment guidance. Content was also cross-checked against WHO typhoid facts for clinical terminology.

Educational study summary only. This document is not a substitute for professional medical diagnosis or treatment.